Estimating Social Returns from Liberalisation

Submitted for DSE - Introduction to Causal Inference

By

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Abstract

This paper evaluates whether India's 1991 economic liberalisation produced significant gains in social development beyond what would have occurred in its absence, providing causal evidence for the Bhagwati-Sen debate. Using the Synthetic Control Method, we construct a weighted combination of 26 developing countries that closely replicates India's pre-reform trajectory in infant mortality and adult literacy over the period 1980-90. Comparing India's post-1991 outcomes to this synthetic counterfactual, I find no evidence of a structural break attributable to liberalisation: the treatment effect gaps are small in magnitude and statistically indistinguishable from placebo gaps generated by assigning the 1991 treatment date to donor countries, with permutation p-values of 0.333 and 0.600 for infant mortality and adult literacy respectively. Leave-one-out robustness checks corroborate this null result. The findings are more consistent with the capability approach advanced by Drèze and Sen (2013) than with the growth dividend hypothesis of Bhagwati and Panagariya (2013), suggesting that market reforms alone were insufficient to alter India's social development path and that the broad improvements observed after 1991 are better attributed to global trends in health and education than to liberalisation per se.

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1 Introduction

Few questions in development economics have generated more sustained controversy than whether economic liberalization alone is sufficient to improve social outcomes, or whether direct state investment in human development is a necessary complement. This debate has particular resonance in the Indian context, where the sweeping market reforms of 1991, often described as the most consequential structural adjustment in the country's post-independence history, coincided with a period of significant, if uneven, improvement in health and education indicators. Yet the central empirical question remains unresolved: were these social gains driven by liberalization's growth dividend, or would they have materialized regardless?

The intellectual fault lines of this debate are captured most sharply in the exchange between Jagdish Bhagwati and Arvind Panagariya on one side, and Jean Drèze and Amartya Sen on the other. Bhagwati and Panagariya (2013) argue that rapid economic growth, catalyzed by liberalization, generates the fiscal revenues that governments require to fund public health and education. On this view, reforms that raise GDP per capita are the most direct path to human development, the so-called 'growth first' position. Drèze and Sen (2013), drawing on Sen's capability approach, respond that growth is a poor substitute for deliberate public action. They point to states like Kerala and Tamil Nadu, which achieved dramatic social progress well before they were rich, as evidence that education, healthcare, and social protection must be actively provided, not merely hoped for as by-products of market expansion.

India's 1991 balance-of-payments crisis and the subsequent liberalization program, encompassing industrial deregulation, trade opening, and financial sector reform, provide a rare natural experiment in which to evaluate these competing claims. If Bhagwati and Panagariya are correct, one would expect the post-1991 period to show a significant structural break in India's social trajectory: the growth dividend should accelerate improvements in infant mortality, literacy, and other indicators of human capability beyond what the pre-existing trend would have predicted. If Drèze and Sen are correct, social outcomes should not diverge sharply from what a credible counterfactual India, one that did not liberalize in 1991, would have achieved.

This paper constructs precisely such a counterfactual using the Synthetic Control Method (SCM), as developed by Abadie and Gardeazabal (2003) and formalized by Abadie, Diamond, and Hainmueller (2010). SCM builds a weighted combination of comparable countries, referred to as a 'synthetic'

India, that closely tracks India's actual social and economic trajectory in the pre-treatment period from 1970 to 1990. The divergence, or lack thereof, between India and its synthetic counterpart after 1991 is then interpreted as the causal effect of liberalization on social outcomes.

The primary outcomes of interest are the Infant Mortality Rate (IMR) and the Adult Literacy Rate. These were chosen because they are the most direct quantitative proxies for the capabilities that Sen places at the center of development - physical health and educational attainment, and because they are the indicators most emphasized by Drèze and Sen (2013) in their critique of India's social record. If liberalization significantly accelerated improvements in both indicators relative to the synthetic counterfactual, the results would lend support to the Bhagwati-Panagariya position. A null or modest treatment effect would instead support the view that market reforms alone were insufficient to alter India's social development path.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature on the effects of liberalization and the application of SCM to related questions. Section 3 describes the data. Section 4 presents the empirical strategy in detail. Section 5 will present results, Section 6 describes limitations and Section 7 will conclude.

2 Literature Review

2.1 The Bhagwati–Sen Debate

The debate between Bhagwati and Panagariya (2013) and Drèze and Sen (2013) is not merely academic, it reflects deep disagreements about the mechanism through which poor countries improve living standards. Bhagwati and Panagariya's central claim is that India's post-1991 growth spurt, directly attributable to liberalization, was the primary driver of poverty reduction and, through the public revenues it generated, of social progress. They point to the sharp deceleration of poverty after 1991 and the expansion of fiscal space that enabled the state to spend more on health and education.

Drèze and Sen (2013) do not deny that growth matters, but they argue that the conversion of income growth into improved human lives is not automatic. It depends on how growth is distributed, how public institutions function, and whether the state actively extends services to marginalized populations. They note that India's social indicators, particularly child malnutrition and female literacy, lagged far behind comparable countries with similar income levels throughout the post-reform period, a pattern inconsistent with a straightforward growth dividend story. This paper aims to bring causal evidence to this debate rather than relying on descriptive comparisons.

2.2 Synthetic Control Method Literature

The Synthetic Control Method was introduced by Abadie and Gardeazabal (2003) in a study of the economic costs of terrorism in the Basque Country, where the authors constructed a weighted combination of other Spanish regions to serve as a counterfactual for Basque GDP in the absence of conflict. The method was subsequently formalized and extended by Abadie, Diamond, and Hainmueller (2010) in an influential study of California's tobacco control program. That paper demonstrated that SCM has key advantages over conventional regression-based approaches when the unit of analysis is a single treated country or region: it makes the counterfactual explicit and observable, it does not require the treated unit to fall within the convex hull of the control units (as difference-in-differences implicitly assumes), and it provides a transparent weighting scheme that can be directly inspected and validated.

A particularly relevant application of SCM in the context of economic reforms is Billmeier and Nannicini (2013), who applied the method to a large global sample of trade liberalization episodes to estimate their effects on GDP per capita. Their key finding is that the effects of liberalization are highly heterogeneous across countries, depending critically on pre-reform institutional quality, human capital, and geographic context. Countries with stronger institutions and higher initial human capital tended to benefit more from opening up. This heterogeneity has direct implications for the Indian case: it suggests that the social returns to liberalization may depend partly on the pre-existing capacity of the Indian state to convert growth into social outcomes.

The method has also been applied to questions of social and health outcomes. Olper, Curzi, and Swinnen (2018) used SCM to study the impact of trade liberalization on child mortality across a sample of developing countries, finding substantial average reductions in under-five mortality following liberalization, though with significant variation across countries. Their finding that mortality reductions were largest in countries with more developed health infrastructure is particularly relevant here, since it suggests that the mechanism linking liberalization to health outcomes may operate through fiscal channels that depend on pre-existing state capacity — precisely the argument Drèze and Sen make.

Grier and Maynard (2016) provide a methodological cautionary note that is directly applicable to this study. In their evaluation of economic policy under Hugo Chávez in Venezuela, they found that while social indicators improved in absolute terms during the Chávez era, they did not significantly outperform the synthetic counterfactual. Their paper illustrates the risk of confusing global trends, in this case, falling mortality and rising literacy across Latin America, with the effects of domestic policy. The same concern applies to India: if social indicators improved globally during the 1990s and 2000s due to technological diffusion, falling drug prices, or international development aid, any comparison that fails to control for these common trends will overstate the effect of liberalization. SCM addresses this problem directly by anchoring the counterfactual to actual countries that shared India's pre-treatment trajectory.

The IMF (2016) applied SCM to a broad set of structural reform episodes, explicitly discussing the identification assumptions required for valid causal inference and the sensitivity of results to donor pool construction. Their methodological discussion is closely followed in this paper, particularly in the approach to selecting and trimming the donor pool.

3 Data

The analysis uses a balanced panel spanning 30 years from 1980 to 2010. This time horizon is chosen to ensure a sufficiently long pre-treatment period (1990-1991) for SCM to identify a synthetic counterfactual that closely tracks India's trajectory, and a sufficiently long post-treatment period (1991-2010) to detect treatment effects that may materialize gradually. The pre-treatment window of 10 years is suitable for what is typically required for SCM to perform reliably (Abadie et al., 2010).

Data are drawn from the World Bank's World Development Indicators (WDI) database provides the social outcome variables and the predictors used to construct the synthetic control.

3.1 Outcome Variables

The two primary outcomes are selected to directly address the capabilities dimension of the Bhagwati–Sen debate.

Infant Mortality Rate (IMR) is defined as the number of deaths among children under one year of age per 1,000 live births in a given year. IMR is a widely used summary statistic for health infrastructure, access to basic medical care, and nutritional conditions. It is sensitive to the kinds of public investments that Drèze and Sen emphasize - sanitation, immunization, maternal health, and is less susceptible than adult mortality to compositional changes in the age structure of the population. IMR data are sourced from the WDI.

Adult Literacy Rate is defined as the percentage of the population aged 15 and above who can both read and write. This variable captures the stock of basic human capital in the economy and reflects the cumulative effect of public education provision over prior decades. It is the indicator most directly referenced by Sen's capability approach as constitutive of human development in itself, not merely instrumental to it. Literacy data are sourced from the WDI.

3.2 Predictor Variables

Following the standard SCM approach outlined in Abadie et al. (2010), the synthetic control is constructed by matching on a set of pre-treatment predictors that are theoretically linked to the outcome variables and that help capture the economic and structural characteristics of the treated unit.

GDP per Capita (PPP) is the primary measure of economic development and captures the fiscal resources available to the state for social spending, the channel emphasized by Bhagwati and Panagariya. It is measured in constant 2017 international dollars from the WDI.

Urbanization Rate is defined as the share of the population living in urban areas, sourced from the WDI. Urbanization affects social outcomes through multiple channels: urban populations typically have greater access to health facilities and schools, but urban density can also increase exposure to communicable disease. Controlling for urbanization ensures that the synthetic control is not confounded by differences in the rural-urban composition of the population.

Trade Openness is measured as total trade (exports plus imports) as a share of GDP, sourced from the WDI. This variable is included as a predictor rather than an outcome because it proxies the intensity of economic integration in the pre-treatment period, helping to ensure that the donor pool countries are structurally similar to India in their openness to international markets prior to 1991.

4 Empirical Design

4.1 The Identification Challenge

Identifying the causal effect of India's 1991 liberalisation on social outcomes faces a fundamental challenge: there is only one India, and no directly observable counterfactual for what its social trajectory would have been in the absence of reform. Standard panel regression approaches address this by using within-unit variation over time or by comparing India to a group of control countries. Both approaches have well-known limitations. Time-series regressions cannot easily separate the effect of liberalisation from other concurrent changes - the debt crisis, the Gulf War, the collapse of the Soviet Union that affected India during the same period. Difference-in-differences estimators require identifying a comparison group that satisfies a parallel trends assumption, which is difficult to justify for a country as economically and demographically distinctive as India.

The Synthetic Control Method, as formalized by Abadie, Diamond, and Hainmueller (2010), offers a transparent solution to this problem. Rather than assuming that any single country or the simple average of a group of countries provides an adequate counterfactual, SCM constructs a weighted combination of control countries, the 'synthetic' India, that is optimized to replicate India's pre-treatment trajectory in the outcome variables and in a set of relevant predictors. The quality of the counterfactual is directly observable from the pre-treatment fit, and the identifying assumption that in the absence of treatment, India would have followed the trajectory of its synthetic counterpart.

4.2 Formal Setup

Following Abadie et al. (2010), let there be $J + 1$ countries indexed by $j = 1, 2, ...J + 1$, observed over T periods indexed by $t = 1, 2, ...T$. Country $j = 1$ is India, the treated unit, which undergoes liberalisation in period $T_0 + 1 = 1991$. Countries $j = 2, 3, ...J + 1$ constitute the donor pool of untreated countries. For each country, let $\hat{Y}^0_{jt}$ denote the potential outcome in the absence of treatment, and $\hat{Y}^1_{jt}$ the potential outcome under treatment. The observed outcome is $\hat{Y}_{jt} = \hat{Y}^0_{jt} + \alpha_{jt} D_{jt}$, where D_{jt} is an indicator equal to one if country j is treated in period t, and α_{jt} is the treatment effect of interest.

Since only India is treated, α_{1t} for $t > T_0$ is the parameter of interest: the effect of liberalisation on India's social outcomes. The problem is that $\hat{Y}^0_{1t}$, i.e., what India's social outcomes would have been after 1991 in the absence of liberalisation, is unobserved. SCM estimates $\hat{Y}^0_{1t}$ as a weighted average of the outcomes of the J donor countries:

$$\hat{Y}^0_{1t} = \sum_{j=2}^{J+1} W_j Y_{jt}$$

where $W = (w_2, w_3, ...w_{J+1})$ is a vector of non-negative weights summing to one. The optimal weight vector W^* is chosen to minimize the pre-treatment discrepancy between India and the synthetic control, subject to the constraints that all weights are non-negative and sum to one. Formally, the weights are chosen to minimize a weighted distance $|X_1 - X_0 W|$, where X_1 is a vector of pre-treatment characteristics for India and X_0 is the corresponding matrix for the donor pool countries.

4.3 Donor Pool Construction

A critical step in any SCM application is the construction of the donor pool. The donor pool must consist of countries that were not treated, i.e., that did not undergo comparable liberalisation during the period used to construct the synthetic counterfactual. Including countries that themselves liberalised during India's treatment window would contaminate the counterfactual, since their post-liberalisation outcomes would be partly the result of their own reforms rather than an unperturbed baseline.

The donor pool in this paper is constructed through principled subjective exclusion, following the guidance of Abadie et al. (2010), who explicitly endorse this approach as appropriate when algorithmic classification introduces noise or when the researcher has well-founded substantive reasons to exclude particular units.

Countries are excluded from the donor pool on one or more of the following grounds. First, countries that underwent significant trade liberalisation or comparable structural adjustment between 1980 and 2000 are excluded, as their post-reform trajectories would not provide uncontaminated counterfactual outcomes. Second, countries are excluded if they are so structurally dissimilar from India, in terms of population size, level of development, or geographic context, that including them in the synthetic control would produce a counterfactual that is not credibly representative of what India might have looked like. Abadie et al. (2010) note that the synthetic control should be constructed from countries that are plausible comparators for the treated unit, and that including very different units can worsen rather than improve the quality of the counterfactual. Third, countries with severe data gaps over the 1980-2010 panel are excluded to ensure that the optimization is not unduly influenced by interpolated values.

Following this process, the donor pool is drawn from developing countries in South Asia, Southeast Asia, sub-Saharan Africa, and the Middle East and North Africa that remained largely closed to trade during the pre-treatment window and for whom high-quality data exist over the full panel. Crucially, the SCM algorithm is then left to assign weights to these countries freely - no further restrictions are placed on the weight vector beyond the non-negativity and summation constraints. Countries that do not contribute to a good pre-treatment fit will receive weights of zero by construction, since the optimization will not assign positive weight to countries that pull the synthetic control away from India's pre-treatment trajectory.

The donor pool comprises 26 developing countries drawn from five regions, listed in Table 1. The regional spread is deliberate: by drawing comparators from South Asia, Southeast Asia, Latin America, the Middle East and North Africa, and Sub-Saharan Africa, the pool ensures that the synthetic counterfactual is not unduly shaped by any single region's idiosyncratic trends in health or education during the sample period.

The South Asian countries (Bangladesh, Nepal, Pakistan, and Sri Lanka) are the most structurally proximate to India in terms of initial income levels, demographic conditions, and colonial institutional heritage. Their inclusion is natural but their weight in the synthetic control is constrained by the requirement that the donor pool consist only of countries that did not undergo comparable liberalisation during India's treatment window. The Southeast Asian group (Indonesia, the Philippines, Thailand, and Vietnam) provides comparators that were at broadly similar stages of development in 1970 but followed distinct policy trajectories. The Latin American countries span a range of income levels and institutional contexts and serve primarily to anchor the GDP per capita dimension of the predictor match. The MENA group (Algeria, Egypt, Jordan, Morocco, and Tunisia) offers comparators with similar initial social indicators and state-led development models. The Sub-Saharan African group (Cameroon, Côte d'Ivoire, Ghana, Senegal, and

South Africa) contributes comparators at lower income levels that help the optimisation match India's pre-treatment outcome trajectory. Turkey is included as a middle-income comparator with a large and heterogeneous economy.

The final set of weights assigned to these countries by the SCM optimisation is reported in Table 3.

Table 1: Donor Pool Composition

Country	Region	ISO	Country	Region	ISO
Bangladesh	South Asia	BD	Algeria	MENA	DZ
Nepal	South Asia	NP	Egypt	MENA	EG
Pakistan	South Asia	PK	Jordan	MENA	JO
Sri Lanka	South Asia	LK	Morocco	MENA	MA
Indonesia	Southeast Asia	ID	Tunisia	MENA	TN
Philippines	Southeast Asia	PH	Cameroon	Sub-Sah. Africa	CM
Thailand	Southeast Asia	TH	Côte d'Ivoire	Sub-Sah. Africa	CI
Vietnam	Southeast Asia	VN	Ghana	Sub-Sah. Africa	GH
Bolivia	Latin America	BO	Senegal	Sub-Sah. Africa	SN
Brazil	Latin America	BR	South Africa	Sub-Sah. Africa	ZA
Colombia	Latin America	CO	Turkey	Other	TR
El Salvador	Latin America	SV			
Mexico	Latin America	MX			
Paraguay	Latin America	PY			
Peru	Latin America	PE			

Notes: 26 countries total across five regions. MENA = Middle East and North Africa

4.4 Pre-Treatment Fit and Covariate Balance

The validity of SCM rests on the quality of the pre-treatment fit. If the synthetic control closely tracks India's actual trajectory in IMR and literacy from 1980 to 1990, then the assumption that it provides a credible post-1991 counterfactual is more plausible. If the pre-treatment fit is poor, the SCM estimate is unreliable regardless of what is found in the post-treatment period. Pre-treatment fit is assessed both visually, by plotting India's actual outcome series against the synthetic control over the full pre-treatment period, and quantitatively, by computing the pre-treatment root mean squared prediction error (RMSPE). A low RMSPE relative to the donor pool countries' pre-treatment errors indicates that the synthetic control provides a better fit for India than any individual donor country does. These are standard benchmarks for assessing SCM quality (Abadie et al., 2010).

In addition to outcome-based fit, the balance of pre-treatment predictor values between India and the synthetic control is reported. This is important because a synthetic control that matches India's pre-treatment outcomes through a combination of countries with very different predictor values might achieve numerical fit for spurious reasons. Ideally, the synthetic control should match India's pre-treatment outcomes and be similar in the underlying economic and demographic characteristics that generate those outcomes.

4.5 Inference

Standard hypothesis testing is not applicable in SCM because the sample consists of a single treated unit, making asymptotic approximations inappropriate. Instead, following Abadie et al. (2010), inference is conducted using a permutation-based placebo test. The SCM procedure is applied iteratively to each country in the donor pool, treating it as if it were the treated unit and constructing a synthetic control from the remaining countries. This generates a distribution of 'placebo' treatment effects which are the gaps between each donor country's actual trajectory and its synthetic counterfactual that can be used to assess whether India's post-1991 gap is unusually large relative to what would be expected by chance.

If India's post-treatment gap is large relative to the distribution of placebo gaps, this constitutes evidence that the observed divergence is unlikely to have arisen by chance and can be attributed to the effect of liberalisation. Countries whose pre-treatment RMSPE is more than twice India's are excluded from the placebo distribution, following standard practice, to ensure that the inference is not dominated by countries for which SCM could not achieve adequate pre-treatment fit (Abadie et al., 2010).

The ratio of the post-treatment RMSPE to the pre-treatment RMSPE is used as the test statistic. A high ratio for India relative to the distribution of placebo ratios constitutes stronger evidence of a treatment effect since it indicates that the post-1991 gap is large not only in absolute terms but also relative to the noise in the pre-treatment period.

4.6 Robustness Checks

Robustness checks are conducted to assess the sensitivity of the main results. The SCM is re-estimated excluding the highest-weighted donor countries one at a time (leave-one-out analysis), to verify that the results are not driven by any single country in the synthetic control.

5 Results

5.1 Predictor Balance and Pre-Treatment Fit

Before interpreting the post-treatment gaps, it is necessary to assess whether the synthetic control provides a credible pre-treatment approximation of India. Table 2 reports the pre-treatment predictor means for India, the synthetic control, and the unweighted donor sample mean for both outcome variables.

Table 2: Predictor Balance: India vs Synthetic India (Pre-Treatment Means)

	India	Synthetic India	Donor Sample Mean
Panel A: Infant Mortality Rate			
GDP per capita (PPP)	2203.15	3668.43	6825.19
Urbanisation rate	24.61	27.46	42.18
Trade openness	13.96	41.35	47.15
IMR (1980)	108.20	108.16	86.19
IMR (1983)	99.80	99.94	76.92
IMR (1985)	95.20	95.25	71.10
IMR (1987)	90.80	90.69	65.64
IMR (1990)	84.40	84.39	59.10
Panel B: Adult Literacy Rate			
GDP per capita (PPP)	2203.15	2237.73	6825.19
Urbanisation rate	24.61	21.69	42.18
Trade openness	13.96	30.97	47.15
Literacy (1980)	40.76	40.77	63.20
Literacy (1983)	42.25	42.25	64.26
Literacy (1985)	43.74	43.74	65.22
Literacy (1987)	45.24	45.23	66.16
Literacy (1990)	47.47	47.51	67.89

Notes: Predictor means are averaged over the pre-treatment period 1980-1990. GDP per capita is measured in constant 2017 international dollars (PPP). The donor sample mean is the unweighted average across all countries in the donor pool. Pre-treatment MSPE: IMR = 8.95×10^{-5}; Literacy = 1.92×10^{-5}

The pre-treatment fit is strong for both outcomes. The lagged values of the outcome variables, which capture the trajectory that the optimisation is most directly asked to replicate, match India almost exactly in both panels. For IMR, the largest discrepancy across the five matched years is 0.14 points (1983); for literacy, it is 0.03 points (1990). This near-exact replication of the pre treatment outcome path gives substantial confidence in the synthetic counterfactual.

The match on structural predictors is less precise. GDP per capita is notably higher in the synthetic control for the IMR specification (3,668 vs. 2,203), though the gap is smaller for the literacy specification (2,238 vs. 2,203). Trade openness is substantially higher in the synthetic than in India for both specifications, reflecting the fact that several donor countries were more open than India even before liberalisation. Both discrepancies are consistent with what Abadie et al. (2010) describe as the inherent tension between matching on predictors and matching on pre-treatment outcomes; because the latter is given greater weight in the optimisation, and because achieving near-exact outcome balance is the precondition for a valid counterfactual, these predictor gaps do not invalidate the analysis. They do, however, warrant interpretive caution. Crucially, both India and the synthetic control sit well below the unweighted donor sample mean on GDP per capita and trade openness, confirming that the optimised synthetic is far closer to India's structural profile than a simple average of the donor pool would be.

5.2 Synthetic Control Weights

Table 3 reports the country weights assigned by the SCM optimisation for each outcome.

For the IMR specification, the synthetic India is constituted primarily by Bangladesh (0.364), Côte d'Ivoire (0.340), and Vietnam (0.171), which together account for approximately 88% of the total weight. For literacy, Nepal (0.592) and Ghana (0.225) are the dominant contributors, with Vietnam again receiving a non-trivial weight (0.166). The concentration of weight on a small number of countries, particularly the prominent role of Vietnam in both specifications, is consequential for the leave-one-out robustness analysis discussed in Section 5.5.

Table 3: Synthetic Control Weights

Panel A: IMR			*Panel B: Adult Literacy Rate*		
Country	ISO	Weight	Country	ISO	Weight
Bangladesh	BD	0.364	Nepal	NP	0.592
Côte d'Ivoire	CI	0.340	Ghana	GH	0.225
Viet Nam	VN	0.171	Vietnam	VN	0.166
Nepal	NP	0.038	Pakistan	PK	0.002
El Salvador	SV	0.023	Morocco	MA	0.002
Cameroon	CM	0.005	Tunisia	TN	0.001
Brazil	BR	0.004	Côte d'Ivoire	CI	0.001
Morocco	MA	0.004	Egypt	EG	0.001
Bolivia	BO	0.004	Cameroon	CM	0.001

Notes: Only countries receiving weight > 0.001 are shown. Remaining donor countries receive zero or negligible weight and do not contribute to the synthetic counterfactual

5.3 Main Results: Post-Treatment Gaps

Figure 1 plots India's actual trajectory against the synthetic counterfactual for both outcomes over the full 1980-2010 period. The pre-treatment fit is visually tight for both outcomes, consistent with the quantitative balance reported above. After 1991, India tracks its synthetic counterpart closely in both panels. There is no sharp divergence at the treatment date and no widening gap in subsequent years.

Figure 1: Infant Mortality Rate

India vs Synthetic India, 1980–2010

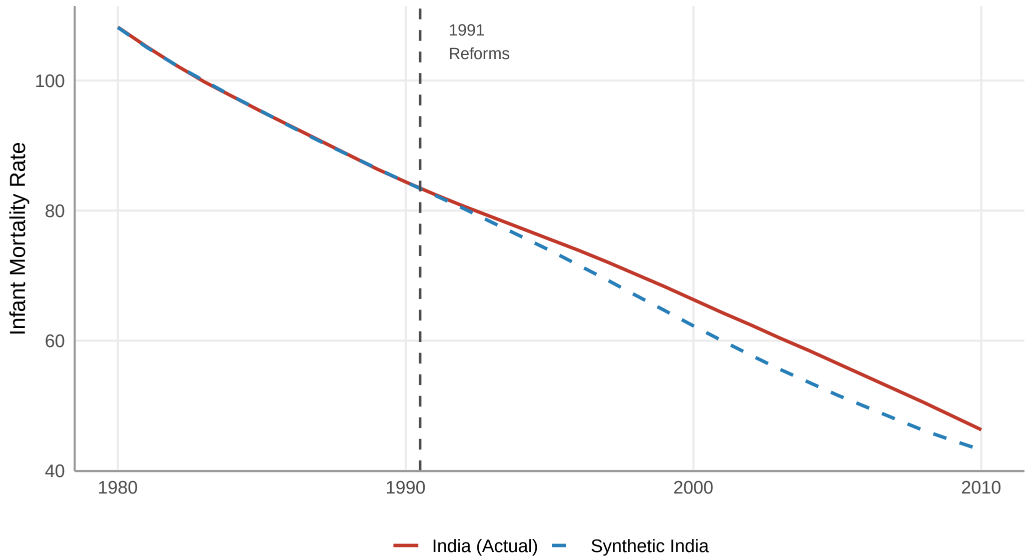


Figure 1: Adult Literacy Rate

India vs Synthetic India, 1980–2010

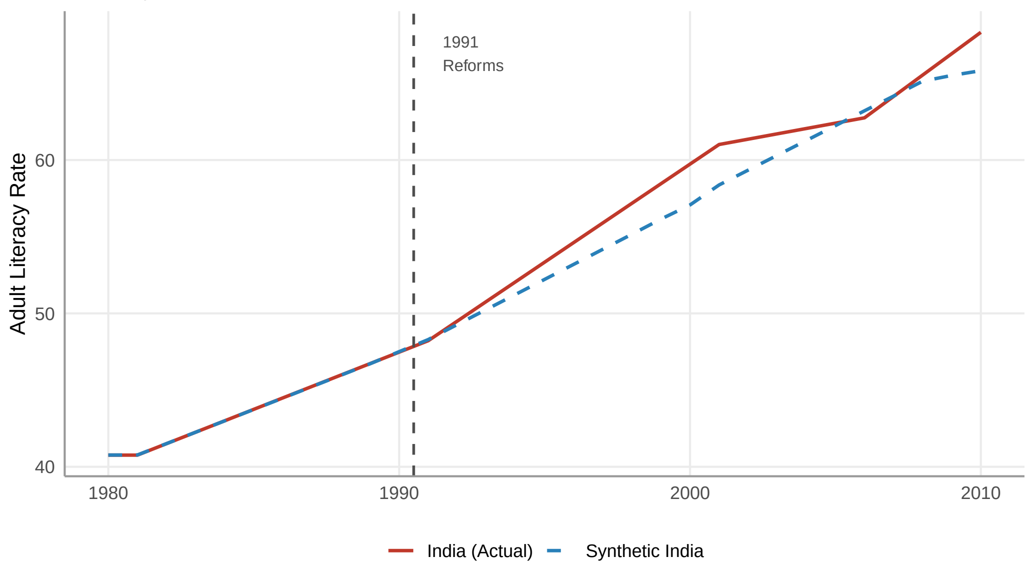
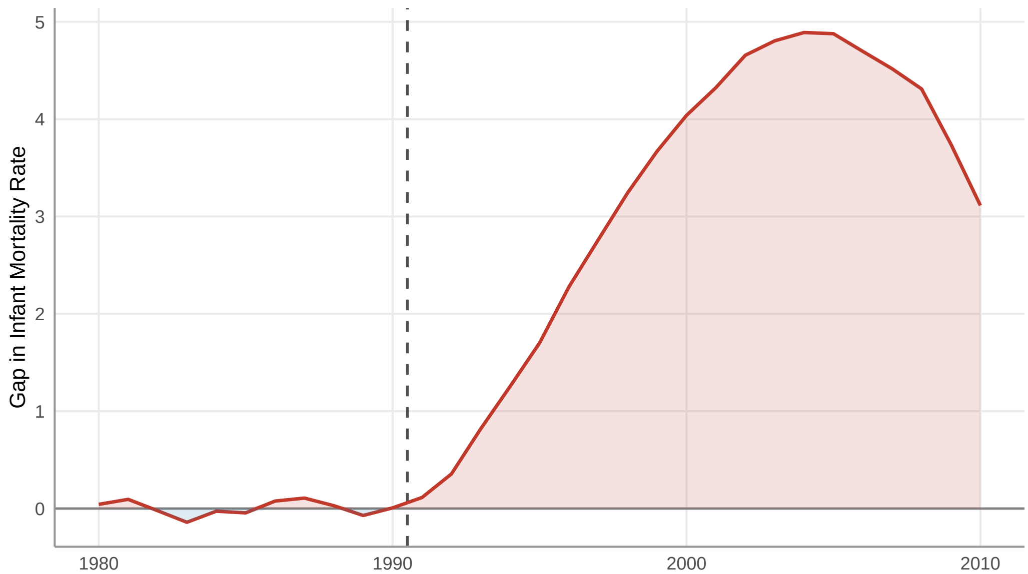
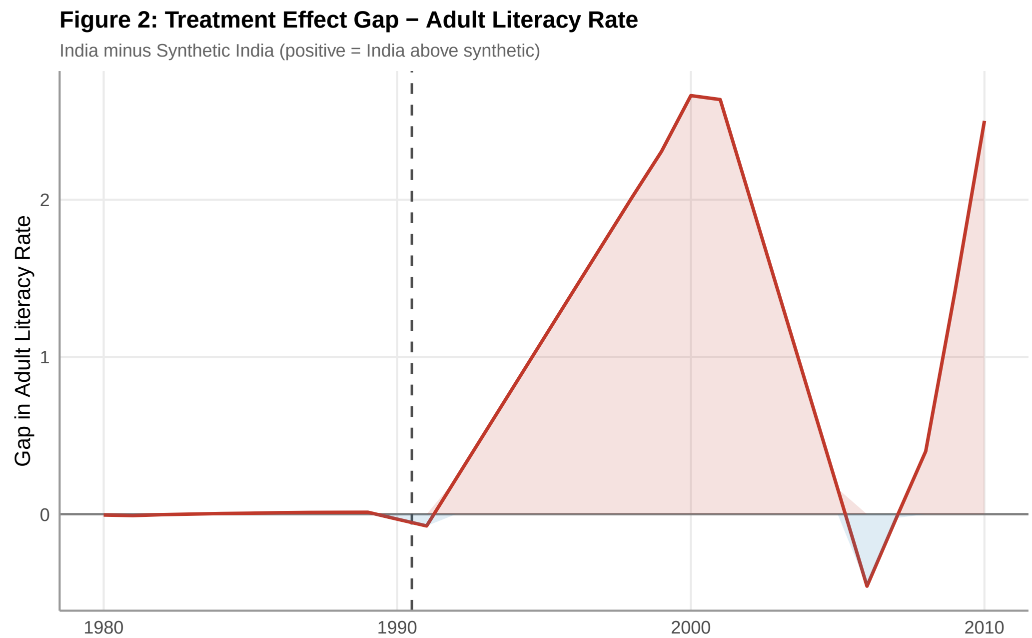


Figure 2 shows the treatment effect gap which is India's actual outcome minus the synthetic counterfactual over time. For IMR, the post-treatment gap i positive and small, reaching approximately 5 points at its peak before converging back toward zero by the end of the panel. A positive gap for IMR indicates that India's mortality rate was higher than the synthetic counterfactual would predict, implying that, if anything, liberalisation was associated with a slight deterioration relative to the counterfactual.

Figure 2: Treatment Effect Gap − Infant Mortality Rate

India minus Synthetic India (positive = India above synthetic)





For literacy, the gap is similarly small and positive, with India's literacy rate marginally exceeding the synthetic after 1991 but the divergence never exceeds approximately 2 percentage points and shows no sustained upward trend.

5.4 Inference

Table 4 reports the RMSPE-based inference statistics.

Table 4: RMSPE-Based Inference

Outcome	RMSPE (Pre)	RMSPE Ratio (Post/Pre)	p-value
IMR	0.072	49.4	0.333
Adult Literacy Rate	0.012	123.0	0.600

Notes: The permutation p-value is the proportion of donor-country placebos whose post/pre RMSPE ratio is at or above India's. Placebos with pre-treatment RMSPE exceeding twice India's are excluded from the inference distribution, following Abadie et al. (2010). With 26 donor countries the minimum achievable p-value is $1/26 \approx 0.038$

Inference in the SCM framework does not rely on asymptotic theory or classical hypothesis tests. Instead, as formalised by Abadie et al (2010), inference is permutation-based: the SCM is applied iteratively to each donor country as a placebo treated unit, and the resulting distribution of post/pre RMSPE ratios serves as the null distribution against which India's ratio is ranked. This procedure is valid regardless of sample size and requires no distributional assumptions, which makes it the appropriate mode of inference when the unit of analysis is a single country.

These p-values must, however, be interpreted with the granularity constraint in mind. With 26 donors, the permutation distribution has only 26 mass points, so the minimum achievable p-value is $1/26 \approx 0.038$. The p-value for IMR is 0.333, meaning that approximately 8 of the 26 donor placebos produce a post/pre RMSPE ratio as large as or larger than India's. For literacy, the corresponding figure is 0.600, or roughly 15 placebos. Neither result provides evidence that India's post-treatment gap is unusual relative to what would be produced by chance under the null of no treatment effect.

Figure 3 illustrates this graphically: India's post-treatment gap (red line) does not stand out from the distribution of placebo gaps (grey lines) for either outcome, and for literacy it sits well within the centre of the placebo distribution.

Figure 3: Placebo Gaps − Infant Mortality Rate

Red = India; Grey = donor placebos (pre−RMSPE = 2x India's). p = 0.333

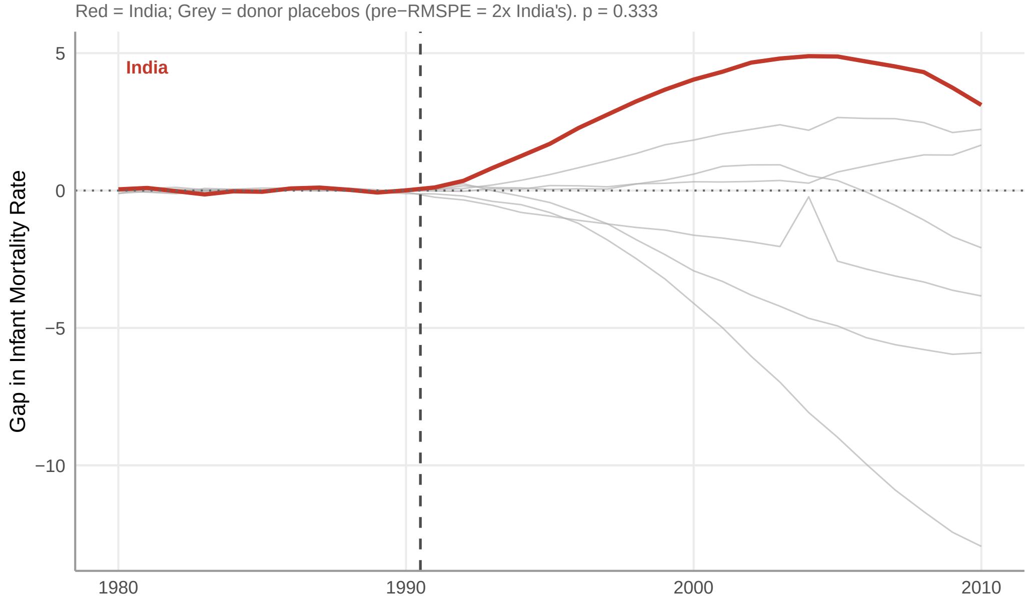


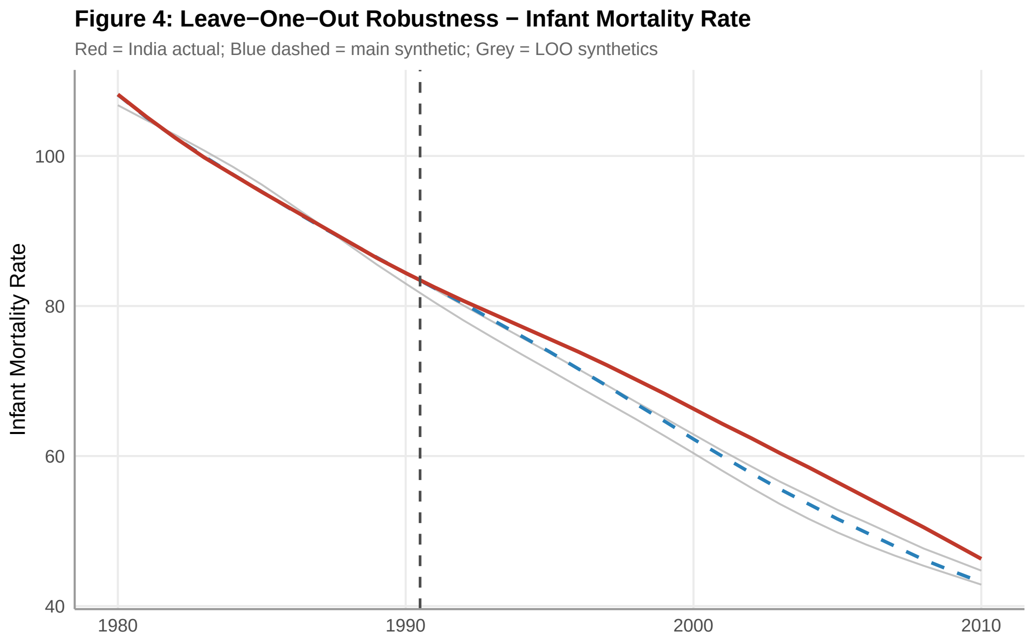
Figure 3: Placebo Gaps − Adult Literacy Rate

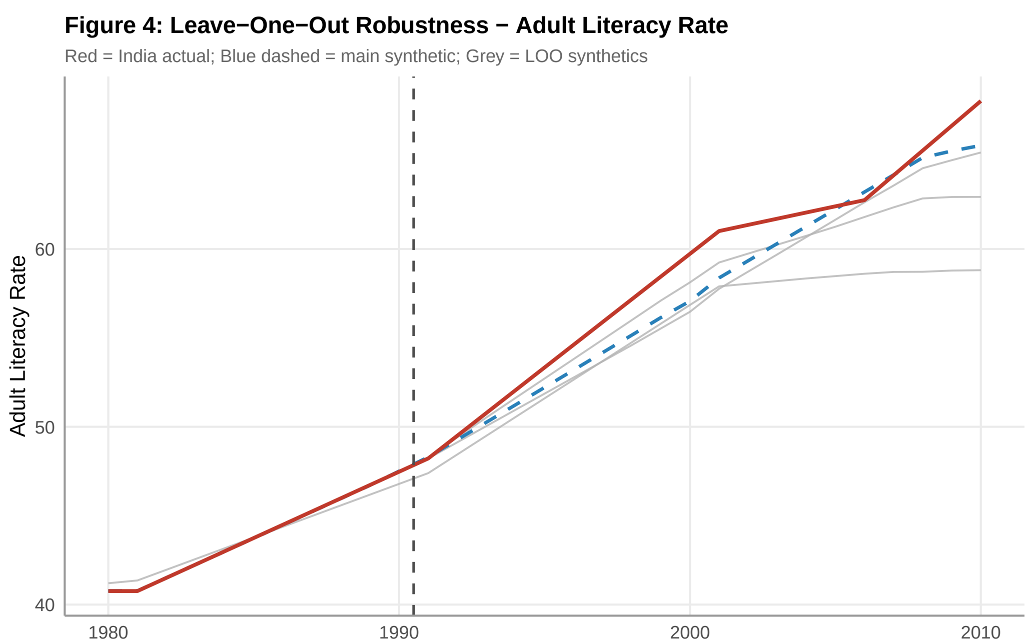
Red = India; Grey = donor placebos (pre−RMSPE = 2x India's). p = 0.6

The high RMSPE ratios (49.4 and 123.0) should not be misread as evidence of a large treatment effect. They are large because the pre-treatment RMSPE values are very small which is a direct consequence of the tight pre-treatment fit making even modest absolute post-treatment discrepancies appear large in ratio terms. The permutation test benchmarks India's ratio against the full placebo distribution and thereby accounts for this scaling correctly.

5.5 Robustness

Figure 4 presents the leave-one-out (LOO) robustness analysis. Across all feasible LOO iterations, the synthetic lines cluster closely around the main synthetic counterfactual, and all remain in close proximity to India's actual trajectory. This confirms that no single donor country is driving the main resuln. The finding of a negligible treatment effect is not an artifact of the particular weight assigned to any one country. The LOO iteration dropping Vietnam was not feasible for the IMR specification as the optimiser could not converge to a finite solution with the remaining donor pool reflecting the substantial weight assigned to that country and the limited size of the donor pool. This infeasibility is itself an informative result: it implies that the synthetic India for IMR relies heavily on Vietnam as a donor, which should be borne in mind when interpreting the counterfactual.





6 Limitations

Several limitations of this study should be acknowledged when interpreting the results.

Counterfactual dependence on high-weight donors. The synthetic control is dominated by a small number of countries - Bangladesh, Côte d'Ivoire, and Vietnam for IMR, and Nepal and Ghana for literacy, despite a donor pool of 26 countries. This concentration reflects the SCM optimisation assigning near-zero weight to countries that do not contribute to pre-treatment fit, and is not a product of donor pool thinness per se. However, it does mean that the counterfactual is sensitive to the characteristics of these specific comparators. The infeasibility of the Vietnam leave-one-out iteration for the IMR specification illustrates this: with 25 remaining donors, the optimiser could not replicate India's pre-treatment trajectory without Vietnam, indicating that no other country in the pool occupies a sufficiently similar position in the predictor space. This dependence is worth acknowledging when assessing the generalisability of the counterfactual, though it does not invalidate it as a concentrated synthetic control is a legitimate outcome of the SCM when few countries genuinely resemble the treated unit.

Predictor imbalance. As noted in Section 5.1, GDP per capita and trade openness are not well matched in the IMR specification. If these structural differences are predictive of the post-treatment evolution of the outcome, the synthetic counterfactual may be a biased estimate of what India would have achieved in the absence of reform. The direction of this bias is uncertain: if higher GDP per capita in the synthetic control translates into faster mortality decline, the counterfactual may be too optimistic, potentially leading to an underestimate of any positive effect of liberalisation. Conversely, if higher trade openness in the synthetic control reflects a degree of market integration that is itself associated with faster social progress, the counterfactual may set too high a benchmark.

Outcome coverage. This study examines only two social indicators - infant mortality and adult literacy. These were chosen for their direct correspondence to the capability approach and for data quality, but they do not capture the full range of social outcomes that either Bhagwati and Panagariya or Drèze and Sen emphasise. Maternal mortality, child malnutrition, secondary school enrollment, and access to sanitation are all relevant dimensions that may respond differently to liberalisation. A more comprehensive assessment would examine a wider set of indicators.

The 1991 reform as a unitary treatment. The SCM framework treats the 1991 liberalisation as a single, discrete treatment event. In practice, India's reform process was gradual and multidimensional. Trade liberalisation, industrial deregulation, and financial sector reform proceeded at different speeds and were implemented partially reversed and re-deepened across subsequent governments. The binary treatment classification therefore abstracts from considerable within-period heterogeneity in the intensity of reform, which means that the estimated treatment effect should be interpreted as an average effect of the broad reform programme rather than of any specific policy component.

Concurrent shocks. The post-1991 period coincided with several global developments sich as the diffusion of oral rehydration therapy and vaccines, the expansion of international aid for primary education, and the broad decline in child mortality observed across the developing world, that affected India and its donor pool simultaneously. While the SCM framework controls for such common trends by construction, to the extent that India was differentially exposed to these shocks relative to the synthetic control, the estimated gap would conflate liberalisation effect with the effects of other concurrent changes.

7 Conclusion

This paper used the Synthetic Control Method to evaluate the social returns to India's 1991 economic liberalisation, asking whether the reforms produced a significant acceleration in infant mortality reduction and literacy accumulation beyond what a credible counterfactual India would have achieved.

The central finding is that they did not. For both outcomes, the gap between India's actual trajectory and the synthetic counterfactual is small in magnitude and statistically indistinguishable from the placebo distribution, with permutation p-values of 0.333 and 0.600 for IMR and literacy respectively. The leave-one-out robustness checks corroborate this null result.

These findings are more consistent with the position advanced by Drèze and Sen (2013) than with the Bhagwati and Panagariya (2013) growth dividend hypothesis. If the fiscal resources generated by post-1991 growth were the primary engine of social progress, one would expect a structural break in India's social trajectory after 1991 that clearly outperforms the counterfactual. No such break is evident in the data. India's infant mortality and literacy rates after 1991 closely track the levels that a weighted combination of comparable non-liberalising countries would have predicted, suggesting that the broad social trends of the period, driven by technological diffusion, global health initiatives, and long-run demographic transitions, account for most of the observed improvement, rather than the specific stimulus of liberalisation.

This is not to say that liberalisation had no effects on social outcomes. The estimates are imprecise, the donor pool is small, and the analysis is limited to two indicators over a particular time horizon. It is possible that liberalisation generated social gains in dimensions not captured here, or that its effects materialised over a longer horizon than the 2010 endpoint of this study permits. The predictor imbalance in the IMR specification — particularly the higher GDP per capita and trade openness of the synthetic control — also means that the counterfactual may set a relatively demanding benchmark, potentially obscuring a modest positive effect.

What the analysis does establish is that the simple claim that market reforms were the primary driver of India's post-1991 social progress is not supported by a credible counterfactual comparison. The gap between India and Synthetic India is too small, and too consistent with what placebo tests would produce by chance, to sustain that interpretation. This finding reinforces the broader lesson of the comparative development literature, illustrated by Billmeier and Nannicini (2013), Olper et al. (2018), and Grier and Maynard (2016), that the social effects of liberalisation are highly context-dependent and cannot be read off from aggregate growth performance alone. For India, as Drèze and Sen argued, the conversion of economic growth into improved human lives appears to require deliberate public action that market forces alone do not reliably deliver.

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